

CLASS ASSIGNMENT

1. Consider the following relational schema

STUDENT_COURSE(Reg_no, Sname, DoB, Address, Crs_Code, Crs_name, Credits, Hours).

Explain, with example, update anomaly, deletion anomaly and insertion anomaly that may be present in a state of the above table.

2. Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies

$F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$.

What is the key for R?

Decompose R into 2NF

3. Consider the following relation R:

R (Doctor_id, Patient_id, Date, Diagnosis, Treat_code, Charge)

In the above relation, a tuple describes a visit of a patient to a doctor along with a treatment code and daily charge. Assume that diagnosis is determined (uniquely) for each patient by a doctor. Assume that each treatment code has a fixed charge (regardless of patient). Is this relation in 2NF? Justify your answer and decompose.

4. Consider the relation REFRIG(Model_no, Year, Price, Manuf_plant, Color), which is abbreviated as REFRIG(M, Y, P, MP, C), and the following set F of functional dependencies:

$F = \{M \rightarrow MP, \{M, Y\} \rightarrow P, MP \rightarrow C\}$

(a) Evaluate each of the following as a candidate key for REFRIG, giving reasons why it can or cannot be a key: $\{M\}$, $\{M, Y\}$, $\{M, C\}$.

(b) Based on the above key determination, state whether the relation REFRIG is in 2NF? Give proper reasons.

(c) Consider the decomposition of REFRIG into $D = \{R_1(M, Y, P), R_2(M, MP, C)\}$. Is this decomposition lossless? Show why.